

Python Fundamentals Quiz

Topic: Python programming fundamentals | Difficulty: Medium | Questions: 5 | Model: gemini-2.5-flash

1. Which of the following Python data types is *immutable*?

- A. list
- B. dict
- C. set
- D. tuple (correct)**

Explanation: Tuples are immutable sequences, meaning their elements cannot be changed after creation. Lists, dictionaries, and sets are all mutable.

2. What will be the output of the following Python code snippet?

```
```python
for i in range(1, 5):
 if i % 2 == 0:
 print(i)
```
```

- A. 1
- 2
- 3
- 4
- B. 2**
- 4 (correct)**
- C. 1
- 3
- D. 0

2
4

Explanation: The `range(1, 5)` generates numbers 1, 2, 3, 4. The `if i % 2 == 0` condition checks for even numbers. So, only 2 and 4 will be printed.

3. Consider the following Python function:

```
```python
def calculate_sum(a, b=10, c=5):
 return a + b * c
```

```
result = calculate_sum(2, c=3)
```

```
```
```

What will be the value of `result`?

- A. 17
- B. 32 (correct)**

C. 20

D. 26

Explanation: When `calculate_sum(2, c=3)` is called, `a` becomes 2. `b` uses its default value of 10. `c` is overridden to 3. The calculation is `2 + 10 * 3`, which is `2 + 30 = 32`.

4. What is the output of the following code?

```
```python
my_list = [1, 2, 3, 4, 5]
print(3 in my_list and 6 not in my_list)
```
```

A. True (correct)

B. False

C. Error

D. None

Explanation: The expression `3 in my_list` evaluates to `True` because 3 is an element of `my_list`. The expression `6 not in my_list` also evaluates to `True` because 6 is not an element. `True and True` results in `True`.

5. What will `my_list` contain after executing the following code?

```
```python
my_list = [10, 20, 30]
my_list.append(40)
my_list.insert(1, 15)
my_list.pop(3)
```
```

A. [10, 15, 20, 30]

B. [10, 15, 20, 40] (correct)

C. [10, 15, 30, 40]

D. [10, 20, 15, 40]

Explanation: 1. `my_list = [10, 20, 30]`

2. `my_list.append(40)` results in `[10, 20, 30, 40]`

3. `my_list.insert(1, 15)` results in `[10, 15, 20, 30, 40]`

4. `my_list.pop(3)` removes the element at index 3 (which is 30), resulting in `[10, 15, 20, 40]`.

Answer Key

1. **D** tuple
2. **B** 2
 4
3. **B** 32
4. **A** True
5. **B** [10, 15, 20, 40]